# Family 7: a global, three-resolution input tensor
for a forward model of the Earth's surface state

blauewelt/earth — design note

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**Abstract**

Family 7 is the first input tensor in this programme to cover the whole planet. It records the ocean, atmosphere, land and ice surface on a 0.25° latitude–longitude grid from pole to pole, in fixed five-day bins from 1982 to the end of 2024, in 54 channels drawn from six public sources. Its distinguishing design choice is that channels are stored at their native resolution in three separate groups rather than in one dense array, and are read at a common 0.25° position by a nearest-cell lookup: a 1.9° reanalysis field stored at 0.25° would be sixty copies of every number. The tensor holds $1.38 \times 10^{10}$ observed values against the $2.5 \times 10^{9}$ of its North Atlantic predecessor, occupies 52.9 GB, and is published with a per-file hash. On the sub-block it shares with that predecessor the two agree to float16 precision on 99.2–99.5% of cells. We state the grid, the channels and their processing rules, the missing-value semantics, the validation, and the decisions that shaped it, so that a reader can load and interpret the files without further documentation.

## 1 Purpose

The programme builds a forward model of the Earth's surface state: encode what is observed at each grid cell into a compact vector, predict that vector forward in time, and read out quantities of interest — Atlantic overturning transport, the El Niño index, sea ice — from the predicted state. Its earlier tensors (families 3–5) covered a North Atlantic window, 0–70° N and 100° W–20° E, at monthly, five-day and daily cadence. Two arguments drove the move to a global tensor. First, the model size is set by the data rather than by compute at this scale: on the North Atlantic the rolled forecast skill was flat from 7.6 M to 206.66 M parameters, and the remedy is more independent samples, which new regimes supply and a longer record cannot. Second, a window edge is a boundary the model must learn as if it were physics; on a sphere there is none.

## 2 Grid and time axis

The horizontal grid is point-aligned at 0.25°: latitudes $-90 + 0.25\,i$ for $i = 0, \ldots, 720$ (south first, 721 rows) and longitudes $-180 + 0.25\,j$ for $j = 0, \ldots, 1439$ (1,440 columns), giving 1,038,240 cells. Longitude wraps: column 1439 (179.75° E) is adjacent to column 0. A coarse 1° grid of $181 \times 360$ points is defined the same way. Every fourth 0.25° point is a 1° point, and a coarse channel is read at a fine position $(y, x)$ by

$$y_1 = \min\bigl(\lfloor y/4 + \tfrac{1}{2} \rfloor,\ 180\bigr), \qquad x_1 = \lfloor x/4 + \tfrac{1}{2} \rfloor \bmod 360, \tag{1}$$

so the two fine points on either side of a coarse point round to it and the last two columns wrap to −180°.

Time is a fixed five-day bin, a *pentad*, counted from the epoch 1982-01-01: bin $b$ covers days $[5b,\ 5b{+}5)$ after the epoch. There are 3,142 bins, the last opening on 2024-12-31. Bins are calendar-free — they do not align with months or years, and a year holds 73 of them — which is deliberate: a bin is an averaging window, not a date, and a model should not be handed a calendar it can memorise.

## 3 Channels

Channels are stored in three groups, each an array of shape $[T, H, W, C]$ in little-endian float16, C order, with a 128-byte NumPy header. One time step of one group is therefore a contiguous byte range and can be fetched on its own; the programme's web front end paints a global frame from a single HTTP range read of 14.5 MB.

| group | grid | shape | size | content |
|---|---|---|---|---|
| `g025` | 0.25° | $[3142, 721, 1440, 7]$ | 45.67 GB | ocean surface |
| `g100` | 1° | $[3142, 181, 360, 15]$ | 6.14 GB | atmosphere and land, every surface |
| `rg100` | 1°, monthly | $[252, 181, 360, 32]$ | 1.05 GB | ocean interior, Argo |

Table 1: The three channel groups. `rg100` has one row per month (2004-01 to 2024-12); each is assigned to the pentad containing that month's 15th, and the list of live bins ships with the tensor.

### 3.1 Ocean surface, `g025`

Five channels come from the GLORYS12 ocean reanalysis [1]: current speed, eastward and northward current, sea-surface height and the $\log_{10}$ of mixed-layer depth, each a pentad mean of daily fields with at least three days present. Speed and the two components are written in one pass so that $\sqrt{u^2+v^2}$ equals the stored speed exactly. GLORYS covers 80° S northward and begins in 1993, so rows 0–39 and bins 0–802 of these channels are absent. Two channels are observations rather than reanalysis: sea-surface temperature from OISST v2.1 [2], bilinearly interpolated to the point grid and averaged over the pentad, present from 1982; and sea-ice concentration from the same product's `icec` field. The sea-surface temperature channel is deliberately *not* filled over land or ice from any model (§6).

### 3.2 Atmosphere and land, `g100`

Fifteen channels come from NCEP/NCAR Reanalysis 1 [3] on its T62 Gaussian grid ($\approx 1.9°$), bilinearly interpolated to the 1° point grid and averaged over the pentad: wind stress on the surface (eastward and northward, sign-flipped from the file's momentum flux) and the within-pentad standard deviation of each; 2 m air temperature; 10 m wind components; surface pressure; precipitation rate and snow water equivalent, each stored as $\log(1+x)$; volumetric soil moisture and soil temperature in the top 10 cm, masked to land *before* regridding so the coastal interpolation renormalises correctly; latent and sensible heat flux with the reanalysis sign convention (positive upward); and skin temperature, defined over every surface.

### 3.3 Ocean interior, `rg100`

Thirty-two channels are the Roemmich–Gilson Argo product [4]: temperature and salinity at sixteen pressure levels from 10 to 1900 dbar, bilinearly interpolated from the product's 1° cell-centred grid and set to missing outside its latitude band (about 64.5° S to 79.5° N) rather than extrapolated toward the poles.

### 3.4 Normalisation

Every channel is standardised before the float16 write, $x' = (x-\mu)/\sigma$ with $(\mu, \sigma)$ computed over all finite values in all bins, and the pairs are stored per group. A stored value is therefore in standard deviations, not in the channel's unit; float16 resolves a standardised value near 1 to about $10^{-3}\,\sigma$, which is 0.01°C for sea-surface temperature and $2 \times 10^{-4}\,\mathrm{m\,s^{-1}}$ for currents. Anomalies are not stored: the training code derives them at load time from a training-years-only climatology.

## 4 Missing values

`NaN` means "not observed here at this time" and is information rather than damage. The pattern is systematic. Ocean channels are absent over land; GLORYS channels are absent south of 80° S and before 1993; soil channels are absent over sea; the Argo group is absent outside 2004–2024, outside its latitude band, and in any bin not on its monthly list. One case is easy to misread: **sea-ice concentration is absent wherever it is below 15%**, because that is how the source reports ice. Measured on the published bytes for the bin opening 2015-01-03, 155,141 of 1,038,240 cells are finite, every one in $[0.150, 1.000]$, none zero; a mid-Atlantic cell reads a sea-surface temperature of 21.3°C and no ice value. A consumer wanting zeros on open water sets the channel to zero where sea-surface temperature is finite and ice is not. Overall, $1.069 \times 10^{10}$ of the ocean group's $2.28 \times 10^{10}$ slots are finite; the three groups together hold $1.38 \times 10^{10}$ observed values.

# 5 Statics and truth series

Two static fields ship with the tensor. A surface class per 0.25° cell — 0 ocean (702,642 cells), 1 land (226,495), 2 ice sheet or glacier (107,074, from Natural Earth's glaciated-area polygons [6]), 3 inland water (2,029, from its lake polygons) — with ice taking precedence over ocean where both apply, because the surface an instrument sees on an ice shelf is ice. And elevation in metres from ETOPO 2022 [5], averaged over the cell, negative under the sea. Two truth series are carried as (bin, value) pairs: the RAPID array's overturning transport at 26.5° N in Sverdrups, 1,459 pentads from 2004 to 2024, and the Florida Current cable transport, 2,490 pentads from 1982. Both are Atlantic; they exist so that a global model is scored on the same instrument as its predecessors.

# 6 Design decisions

**Native resolution, three groups.** The alternative, one dense $[3142, 721, 1440, C]$ array, would store each 1.9° reanalysis value about sixty times and reach 425 GB at the channel count the next phase needs. The cost of the chosen layout is a lookup at read time; the programme's samplers already read coarse channels at the nearest coarse cell, so the lag-0 $3 \times 3$ patch of a fine position serves the same coarse value nine times, which is the model's honest view of a coarse field rather than a rounding artefact.

**Two temperatures, split by instrument.** An earlier build merged observed sea-surface temperature with reanalysis skin temperature into one channel. The merge was reversed: it spliced an infrared–microwave analysis over the sea onto a reanalysis everywhere else at the coastline, which the model would have had to learn as an instrument boundary. The rule adopted is that a channel is shared across surfaces only when both the measurand and the instrument match. Sea-surface temperature therefore stays observed and absent over land; skin temperature is reanalysis everywhere. The two are different measurements and are stored as such.

**Sea ice and snow stay separate.** A fraction and a water mass cannot be merged into a "frozen fraction" without an invented constant.

**The atmosphere source is a stand-in.** NCEP/NCAR Reanalysis 1 was chosen over ERA5 because it needs no account; ERA5 replaces the `g100` fetch in a later phase with no change of layout.

**Resumable, hash-verified publication.** The build runs as one job resumable per stage, with a per-stage specification digest so that a changed stage discards only its own state, and publishes to a public dataset repository with a per-file SHA-256 that is verified by downloading each file back. Three runs were needed — the first cancelled for the temperature merge above, the second failed on a path error in the truth stage — at a total cost of about $1.9.

# 7 Validation

Ten assertions run before the tensor is trusted, on a CPU with no network: row 0 is the South Pole, asserted on the bytes; $\sqrt{u^2 + v^2}$ equals the stored speed; the absence pattern south of 80° S; the sea-ice unit (the source declares "percent" but its valid range is 0–1, and the reader trusts the range); the surface-class counts; normalisation shapes; truth keys; and resumability across a simulated interruption.

The tensor is not bit-identical to its North Atlantic predecessor, and was not expected to be. On the shared sub-block at the bin opening 2015-01-03, over the 86,698 common ocean cells, currents, sea-surface height and mixed-layer depth agree to two float16 quanta on 99.2–99.5% of cells (correlation

$\geq 0.99995$, median absolute difference $10^{-5}\,\mathrm{m\,s^{-1}}$), with a tail of about 1% differing by up to $0.19\,\mathrm{m\,s^{-1}}$ because the daily fields were binned in a different order at partially masked coastal blocks. Sea-surface temperature agrees to 0.03°C. The global tensor observes 1,756 more coastal cells than the window did, because its regridding renormalises at the coast rather than leaving a masked neighbour empty.

# 8 Limitations

Three limitations are stated rather than hidden. The atmosphere and land channels are a 1.9° reanalysis, the coarsest source in the tensor. Subsurface ocean structure is a monthly climatological product at sixteen levels, present only from 2004 and only within the Argo latitude band. And the record is 43 years: no data source shortens that constraint, and it is the reason the programme's models are small.

# 9 Next steps: the data ladder

The programme's stated aim is a model of $10^9$–$10^{10}$ parameters. At the programme's Chinchilla anchor of one parameter per twenty observed values, that size is earned by $2 \times 10^{10}$–$2 \times 10^{11}$ values, against the $1.38 \times 10^{10}$ family 7 holds. The question is therefore which data to add, in what order, and the answer is set by a measurement rather than by taste: a 0.25° cell is correlated with its neighbours over roughly 300 km and over two to six months at the surface, so the raw value count overstates the independent sample count by about two orders of magnitude, and the flatness of rolled skill from 7.6 M to 206.66 M parameters on the North Atlantic is the symptom. Data that adds *new regimes* or *new physics* is worth its face value; data that adds the same sample five times is not. That ordering is: other spheres first, then depth, then cadence, and horizontal resolution last.

| rung | what is added | values | cost | prerequisite |
|---|---|---|---|---|
| 7 (today) | the global surface at 0.25°, atmosphere and land at 1° | $1.4 \times 10^{10}$ | 53 GB | — |
| L1: the other spheres, properly | ERA5 replacing the NCEP channels at the same layout; observed land-surface temperature (MODIS); scatterometer winds (ASCAT); soil moisture (SMAP, CCI); terrestrial water storage (GRACE); sea-ice concentration and drift (OSI SAF) | $\approx 9 \times 10^{10}$ | ≈170 GB | a Copernicus Climate Data Store account, which only the project owner can create |
| depth | GREP three-dimensional ocean reanalysis at 0.25°: four variables at eight levels, subsurface velocity and abyssal levels the tensor has never had, eleven extra years over Argo | +20 % | +20 GB | existing credentials |
| observed surface overturning | DUACS altimetry (absolute dynamic topography, geostrophic currents) and OSTIA foundation SST, both taken at 0.25° | small | +4.7 GB | existing credentials |
| daily cadence | every group at one-day bins | ×5 raw, $\approx \times 1.5$ indep. | 1.3 TB | a streaming loader; no tensor fits a box |
| the kilometre rung | 1 km daily radiometry, fed to a patch encoder rather than stored as a tensor | $3.5 \times 10^{14}$ | — | the first rung that is compute-bound |

Table 2: The data ladder above family 7, in the order the programme intends to climb it. Values are observed values; "independent" applies the correlation correction of the text.

**Why the spheres come first.** The L1 rung multiplies the observed values by about 4.5 at 170 GB, and every value it adds is new physics: land does not advect, the atmosphere decorrelates in days, ice has its own seasonal cycle. Family 7 already carries the first phase of this rung — the fifteen NCEP channels are the L0 set, with a reanalysis standing in for ERA5 until an account exists — so the layout, the lookup and the loader are in place and the upgrade is a change of source, not of shape. It is also the rung that unlocks the read-outs the programme has promised beyond the Atlantic: a sea-ice head and a land-water head need the observed channels, not the reanalysis ones.

**Why depth comes before pixels.** The overturning is a vertical circulation, and the tensor's only view below the surface is a monthly climatological product. A daily three-dimensional reanalysis at eight levels adds the subsurface velocity field for 20 GB on credentials the programme already holds, and reaches back to 1993 where Argo begins in 2004. It is ranked first in the programme's own data-ladder document on the ratio of new physics to new risk.

**Why resolution comes last.** A 1/12° basin would cost 305 GB for the present channels and, by the correlation argument above, add far fewer independent samples than bytes. The experiment that should precede any such regrid is small: a single 1/12° strip across the RAPID section, 23–30° N, 18 GB for forty channels, asking whether resolution at the section changes the transport read-out at all. If it does not, the basin regrid is never bought. Kilometre-scale radiometry is a different object altogether — $3.5 \times 10^{14}$ values cannot be a tensor — and enters, when it enters, as a patch-encoder input in the style of AlphaEarth's per-source encoders.

**Two cheap items that precede all of the above.** The 2025–2026 continuation of every source adds 4.5% of end-bins nearly for free and lets the terminal holdout run to 2026 instead of 2024. And the static fields the programme downloads but does not yet use — bathymetric slope, the Coriolis parameter and its gradient, distance to coast — are megabytes with the highest information per byte of anything on this list.

# References


[1] Lellouche, J.-M. et al. (2021). The Copernicus Global 1/12° Oceanic and Sea Ice GLORYS12 Reanalysis. *Front. Earth Sci.* 9:698876.

[2] Huang, B. et al. (2021). Improvements of the Daily Optimum Interpolation Sea Surface Temperature (DOISST) Version 2.1. *J. Climate* 34, 2923–2939.

[3] Kalnay, E. et al. (1996). The NCEP/NCAR 40-Year Reanalysis Project. *Bull. Amer. Meteor. Soc.* 77, 437–471.

[4] Roemmich, D. and Gilson, J. (2009). The 2004–2008 mean and annual cycle of temperature, salinity, and steric height in the global ocean from the Argo Program. *Prog. Oceanogr.* 82, 81–100.

[5] NOAA National Centers for Environmental Information (2022). ETOPO 2022 15 Arc-Second Global Relief Model.

[6] Natural Earth, version 5.1.1, 1:10m physical vectors (glaciated areas; lakes). Public domain.